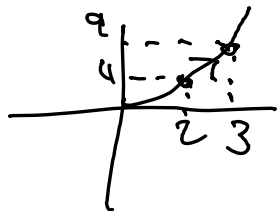


typo for surface integrals (already corrected in notes)
curve descriptions and direction

divergence

divergence theorem

suppose we want to do a line integral
along



$$y = x^2 \text{ with } 2 \leq x \leq 3$$

we'd say let $\vec{r}(t) = (t, t^2)$ $t \in [2, 3]$ interval $2 \leq t \leq 3$

What if I need to go the other way?

You can get away with saying: let $t: 3 \rightarrow 2$ for the purpose of line integrals of vector fields

However, technically that messes w/ our def of a vector fn.

How would you do it properly? We'd make

$$\vec{u}(t) = \vec{r}(a+b-t) \quad \text{eg} \quad (a+b-t, (a+b-t)^2) \quad \text{with}$$

$t \in [a, b]$. Let's see - does it generate the points in the right order? $\vec{u}(t) = (5-t, (5-t)^2) \quad t \in [2, 3]$

$$\vec{u}(2) = (3, 9) \quad \vec{u}(2.5) = (2.5, (2.5)^2) \dots \vec{u}(3) = (2, 4)$$

thus, in general, given a curve C described by $\vec{r}(t) : [a, b] \rightarrow \mathbb{R}^n$. The curve $-C$ denotes the same pts as C , but with opposite orientation, described by $\vec{u}(t) = \vec{r}(a+b-t) \quad t \in [a, b]$.

Now, what happens if I do a line integral along C and $-C$? You'll see in your tut that $\int_C f ds = \int_{-C} f ds$
(add up scalars in any direction)

but $\int_{-C} \vec{F} \cdot d\vec{r} = - \int_C \vec{F} \cdot d\vec{r}$ because dot product!

So the orientation in Green's and Stokes is clearly significant!

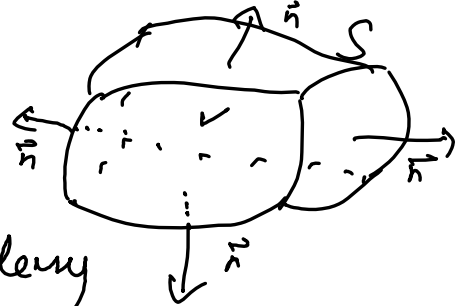
Divergence: Let $\vec{F}: U \rightarrow \mathbb{R}^3$ be a vector field w/
 $\vec{F}(x,y,z) = (P(x,y,z), Q(x,y,z), R(x,y,z)) \quad \forall (x,y,z) \in U$.
 w/ P, Q, R having cont. partial derivs. The
 divergence of the v.f $\vec{F}$ is the scalar field

$$\begin{aligned} \operatorname{div}(\vec{F}) &= \nabla \cdot \vec{F} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \cdot (P, Q, R) \\ &= \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \end{aligned}$$

notice gradient : scalar field $\rightarrow$ vector field
 div : vector field $\rightarrow$ scalar field
 curl : vector field $\rightarrow$ vector field

also: remember: curl cross $\nabla \times \vec{F}$
 div. dot $\nabla \cdot \vec{F}$

\ Divergence Thm (Gauss Thm)



Let V be a closed, bounded region in $\mathbb{R}^3$ with a piecewise smooth boundary surface $S = \partial V$ (denotes bd of region V) with an outward orientation (normals point out). Let $\vec{F} = (P, Q, R) : U \rightarrow \mathbb{R}^3$ is a smooth, 3D vector field defined in a region U containing the bounded region V and its boundary surface S , then

$$\iint_S \vec{F}(x, y, z) \cdot \vec{n}(x, y, z) dS = \iiint_V \operatorname{div}(\vec{F})(x, y, z) dx dy dz$$

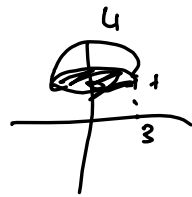
So, if S is a closed surface, we can do the flux integral via a volume int. w/ integrand some derivs of $\vec{F}$

Green's & Stokes' : region: surface
 boundary: curve

Divergence : region: volume
 boundary: surface.

eg. $\vec{F}(x, y, z) = (x, y, z-1)$. S : union of hemisph.
 $x^2 + y^2 + (z-1)^2 = 9$, $1 \leq z \leq 4$, plane $z=1$, $\vec{n}$ is the
 outward unit normal field. Eval $\iint_S \vec{F} \cdot \vec{n} \, dS$

Let's first do this as the volume integral:
 Div. thm says $\iint_S \vec{F} \cdot \vec{n} \, dS = \iiint_V \text{div}(\vec{F}) \, dx \, dy \, dz$



$$\text{div } \vec{F} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z-1) = 3 \quad \text{integrand}$$

$V = 1 \leq z \leq \sqrt{9 - x^2 - y^2} + 1$ x, y : disk rad 3

cylinder: $0 \leq \theta \leq 2\pi$, $0 \leq r \leq 3$, $1 \leq z \leq \sqrt{9 - r^2} + 1$. $\int_{\theta=r}^{\text{integrand}}: 3$

$$\int_0^{2\pi} \int_0^3 \int_1^{1+\sqrt{9-r^2}} 3r \, dz \, dr \, d\theta = \int_0^{2\pi} \int_0^3 3r(1+\sqrt{9-r^2}-1) \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^3 3r\sqrt{9-r^2} \, dr \, d\theta = 2\pi \int_0^3 3r\sqrt{9-r^2} \, dr$$

let $u = 9-r^2$ $du = -2r \, dr \Rightarrow 3r \, dr = \frac{3du}{-2}$

$$= 2\pi \int_{-\frac{3}{2}}^{\frac{3}{2}} \sqrt{u} \, du = 2\pi \int_{-\frac{3}{2}}^{\frac{3}{2}} -\frac{3}{2} u^{1/2} \, du = 2\pi \left[-\frac{3}{2} \cdot \frac{u^{3/2}}{\frac{3}{2}} \right]_{-\frac{3}{2}}^{\frac{3}{2}} =$$

$$-2\pi (9-r^2)^{3/2} \Big|_0^3 = 0 - -2\pi \cdot 9^{3/2} = 2\pi \cdot 27 = 54\pi$$

Now let's compare the surface integral!

Firstly: $S = S_1 \cup S_2$

$$S_1: g(x,y) = 1 + \sqrt{9 - x^2 - y^2} = z$$

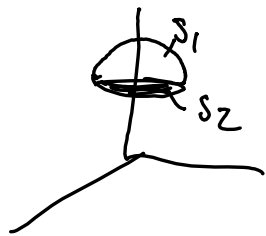
so normal: $(-g_x, -g_y, 1) = \left(\frac{-x}{\sqrt{9-x^2-y^2}}, \frac{-y}{\sqrt{9-x^2-y^2}}, 1 \right)$

$$\vec{F}(x,y,g(x,y)) = (x, y, g(x,y)-1) = (x, y, \sqrt{9-x^2-y^2})$$

$$\iint_{S_1} \vec{F} \cdot \vec{n} dS = \iint_D \underbrace{\vec{F}(x,y,g(x,y)) \cdot (-g_x, -g_y, 1)}_{\text{on surface}} dx dy =$$

$$\iint_D (x, y, \sqrt{9-x^2-y^2}) \cdot \left(\frac{x}{\sqrt{9-x^2-y^2}}, \frac{y}{\sqrt{9-x^2-y^2}}, 1 \right) dx dy = \iint_D \frac{9}{\sqrt{9-x^2-y^2}} dx dy$$

$D: x^2 + y^2 \leq 9$ so $\int_0^{2\pi} \int_0^3 \frac{9}{\sqrt{9-r^2}} r dr d\theta = 54\pi$ (very)



S_z : part of plane $z=1$ for which $x^2+y^2 \leq 9$ 

From the thm, we need $\vec{n}$ to

point out so $g(x,y)=1 \Rightarrow (-g_x, -g_y, 1) = (0,0,1)$
But that points in, so use $(0,0,-1)$.

$$\iint_S \underbrace{\vec{F}(x,y,g(x,y))}_{\text{on plane}} \cdot \vec{n} \, dS = \iint_D (0,0,1-1) \cdot (0,0,-1) \, dx \, dy$$

$$= \iint_D 0 \, dx \, dy = 0 = \int_0^{2\pi} \int_0^3 0 \, r \, dr \, d\theta = 0.$$

check in notes: we can describe the same surface by

$$\vec{r}(s,t) = (s \cos t, s \sin t, 1) \quad 0 \leq s \leq 3 \quad 0 \leq t \leq 2\pi$$